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Reconstruction and Denoising of EEG Signal Using Alternating Direction Method of Multipliers

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***Abstract*—The tremendous progress of big data acquisition and processing in the field of neural engineering has enabled a better understanding of the patient's brain disorders with their neural rehabilitation, restoration, detection, and diagnosis. Electroencephalography (EEG) is a method to record an electrogram of the electrical activity on the scalp that has been shown to represent the macroscopic activity of the surface layer of the brain underneath. However, EEG signals are contained by different noises. This report presents efficient reconstruction and denoising of EEG signal based on Alternating Direction Method of Multipliers. The alternating direction method of multipliers (ADMM) has been widely explored due to its broad applications, and its convergence has been gotten in the real field. This report discusses the alternating direction method of multipliers (ADMM), a simple but powerful algorithm that is well suited to distributed convex optimization, and in particular to problems arising in applied statistics and machine learning. It takes the form of a decomposition-coordination procedure, in which the solutions to small local subproblems are coordinated to find a solution to a large global problem. The algorithm in the report show that the proposed ADMM-based method performs better in EEG-denoising than Compress Sensing (CS) method which is simple and traditional method. Furthermore, the proposed algorithm keeps the detail of the EEG-Signal in reconstruction and achieves smaller root means square error (RMSE) for small-iteration.**

Keywords—EEG, ADMM, convex optimization, compress sensing.

I. Introduction

The Human brain is the most complex organ. Since the dawn of medical science, there have been various experiments and studies conducted to understand this complex system. The EEG is considered a tool that can help us understand it in a much better way.

Researchers have worked on recording, acquiring and transmitting these signals. R. Meenu et al [1] used the idea of compressed sensing for detecting epileptic seizures. T. Moy et al. [2] proposed a CS-based EEG signal acquisition and biomarker-extraction system with flexible, thin-film electronics. The work proposed by K. Zeng et al. [3] used the compressibility calculated using the CS theory in EEG for automatic detection of seizure states. The authors used continuous wavelet transform (CWT) to extract features from time-frequency representation, related to ictal, pre-ictal, and ictal EEG.

The Alternating Direction Method of Multipliers (ADMM) is a very popular approach for decomposing a complex optimization problem into an iterative procedure involving a pair of manageable problems[6]. Being proximal in nature, the two problems can often themselves be decomposed into problems that can be tackled in a parallel manner, making the method especially useful in modern, distributed settings [6]. The method has found wide application, for example, in distributed implementations of regression and classification. Examples of such performance analysis, denoising of Bio signals or Biomedical Imaging, decentralized optimization, etc.[4].

The method is remarkably effective, and even rivals specialized solutions. There are several versions of ADMM, and can be derived from the Douglas-Rachford algorithm[6]. The algorithm offers both practical and theoretical advantages. ADMM runs more efficiently, affording relaxation without requiring additional memory, and under assumptions that can be verified without detouring through duality, converges more generally, allowing for infinite dimensions and inexact steps.

Noise removal is a fundamental problem in signal processing. EEG is a vital biomedical tool for brain disease diagnosis. EEG is a noninvasive technique which is safe, harmless, and a quick method of brain diagnosis [5]. The accuracy of the information extracted from a signal requires proper characterization of waveform morphologies and that they are not contaminated by noise. However, EEG is a fairly weak electric signal, and its amplitude is usually in millivolt level. EEG noises include interference by the instrument itself, Baseline wander, human activities, and other factors in the signal [6].

II. Mathematical Background

A. Dual Decomposition

The major benefit of the dual ascent method is that it can lead to a decentralized algorithm in some cases. Suppose, for example, that the objective f is separable (with respect to a partition or splitting of the variable into sub vectors), meaning that

$$f(x) = \sum_{i=1}^{N} f_i\,(x_i) \tag{1}$$

where x = (x1,..., xN) and the variables $x_i \in R^{ni}$ are subvectors of x.

Partitioning the matrix A conformable as

$$A = [A1 \cdots AN\ ],$$

So,

$$A_x = \sum_{i=1}^{N} A_i x_i \tag{2}$$

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The Lagrangian can be written as

$$L(x,y) = \sum_{i=1}^{N} L_i\,(x_i, y)$$
$$= \sum_{i=1}^{N}(f_i\,(x_i) + y^T A_i x_i - \left(\frac{1}{N}\right) y^T b \qquad (3)$$

Which is also separable in x. This means that the x-minimization step (2) splits into N separate problems that can be solved in parallel.

Explicitly, the algorithm is

$$x_i^{k+1} := \mathrm{argmin}_{xi}\, L_i(x_i, y^k) \qquad (4)$$

$$y^{k+1} := y^k + a^k(Ax^{k+1} - b) \qquad (5)$$

The x-minimization step (4) is carried out independently, in parallel, for each i = 1,...,N. In this case, we refer to the dual ascent method as dual decomposition. In the general case, each iteration of the dual decomposition method requires a broadcast and a gather operation. In the dual update step (5), the equality constraint residual contributions $Ax_i^{k+1} + i1$ are collected (gathered) in order to compute the residual $Ax^{k+1} - b$. Once the (global) dual variable $y^{k+1}$is computed, it must be distributed (broadcast) to the processors that carry out the N individual xi minimization steps (4).

Dual decomposition is an old idea in optimization, and traces back at least to the early 1960s.

B. Methods of Multipliers

Augmented Lagrangian methods[1] were developed in part to bring robustness to the dual ascent method, and in particular, to yield convergence without assumptions like strict convexity or finiteness of f.

The augmented Lagrangian for (1) is

$$L_\rho(x,y) = f(x) + y^T(Ax - b) + \frac{\rho}{2}||Ax - b||_2^2 \qquad (6)$$

where ρ > 0 is called the penalty parameter. The augmented Lagrangian can be viewed as the (unaugmented) Lagrangian associated with the problem

$$\textit{minimize } f(x) + \frac{\rho}{2}||Ax - b||_2^2 \qquad (7)$$
$$\textit{subject to } Ax = b.$$

This problem is clearly equivalent to the original problem (1), since for any feasible x the term added to the objective is zero. The associated dual function is

$$g\rho_y = inf_x L_\rho(x,y) \qquad (8)$$

The benefit of including the penalty term is that g and ρ can be shown to be differentiable under rather mild conditions on the original problem. The gradient of the augmented dual function is found the same way as with the ordinary Lagrangian, i.e., by minimizing over x, and then evaluating the resulting equality constraint residual. Applying dual ascent to the modified problem yields the algorithm

$$x_i^{k+1} := argmin_x\, L_\rho(x, y^k) \qquad (9)$$

$$y^{k+1} := y^k + \rho(Ax^{k+1} - b) \qquad (10)$$

Which is known as the method of multipliers for solving (1).

It is easy to motivate the choice of the particular step size ρ in the dual update (10). For simplicity, we assume here that f is differentiable, though this is not required for the algorithm to work. The optimality conditions for (1) are primal and dual feasibility, i.e.,

$$Ax^* - b = 0, \quad \nabla f(x) + A^T y = 0, \qquad (11)$$

Respectively. By definition, x^{k+1} minimizes $L_\rho(x, y^k)$, so

$$0 = \nabla_x L_\rho(x^{k+1}, y^k)$$
$$= \nabla_x f(x^{k+1}) + A^T(y^k + \rho(Ax^{k+1} - b))$$
$$= \nabla_x f(x^{k+1}) + A^T y^{k+1} \qquad (12)$$

We see that by using ρ as the step size in the dual update, the iterate (x^{k+1}, y^{k+1}) is dual feasible. As the method of multipliers proceeds, the primal residual $Ax^{k+1} - b$ converges to zero, yielding optimality.

Compared to dual decomposition, the method of multiplier converges under much more relaxed conditions bad news: quadratic penalty destroys splitting of the x-update, so can't do decomposition So , we use the Alternating Direction Method of Multiplier for Convex Optimization and eliminate the traditional algorithms.

III. Methodology

A. Understanding EEG Signals and Dataset

To work upon any signal one must be familiar with it. The EEG signals are weak signals that are brain emits. Brain emits this signal all the time, while solving problems, while sleeping or any given time of our life. After years of research and studying this phenomenon researchers have been able to detect and visualize these signals. These signals are beyond the control of humans and hence they show the true characteristics of the human mind and its state. It has an untapped psychological potential. According to a few studies, It is said that the frequency of these transmitted signals can help us identify the person's true mental state and emotions. Hence objectifying the results which were considered subjective till now. This project takes assistance from a very rich EEG signal dataset provided by Physio net. It is a dataset containing EEG signals recorded of various subjects when they were solving mental arithmetic tasks. A single file contains various signals and information, however we select the EEG signal from the file shown in Fig. 1.

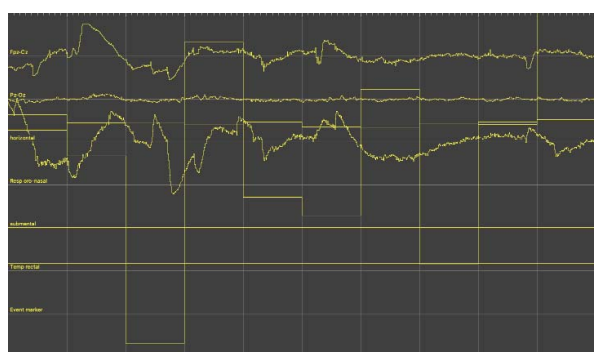

Fig. 1. EEG dataset

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B. Reconstruction

The main objective of the project is to create a model that is able to reconstruct our desired signal from the least possible number of samples. The Nyquist Shannon Theorem states that one requires twice as many samples to reconstruct or visualize a signal without any loss of information. There have been various techniques introduced that defies this core theorem[3]. Compressed sensing which employs convex optimization has been able to deliver many promising results. In our system however we employ another algorithm of Alternate Direction Method of Multiplier(ADMM).

Suppose there exists an EEG signal that is transmitted by the brain which is denoted by 'x' and the observation matrix of the measurement device is given as 'H'. Then the observed signal 'y' can be denoted as shown.

$$y = Hx$$

Now the H is going to be a randomly generated measurement matrix. Clearly the observed signal has much fewer samples compared to the original signal. The task is to recreate the original transmitted signal from these few observed samples. The approach to solve such problems it to initialize an appropriate objective function. Let OBJ(x) be the objective function. It is defined as ;

$$OBJ(x) = D(y, Hx) + \lambda R(x)$$

The R is for regularization and D is the measure of difference between y and x. To further define. The D is as shown.

$$D(y, Hx) = ||y - Hx||_2{}^2$$

The main Regularization is the key difference of ADMM which sets it apart from other algorithms. The R(x) is nothing but an l_1 norm of x and hence OBJ(x) can be defined as.

$$OBJ(x) = ||y - Hx||_2{}^2 + \lambda||x||_1$$

The goal is to minimize the said objective function.

$$J(x) = ||y - Hx|| = (y - Hx)^t(y - Hx)$$

$$= y\hat{}t\, y - 2y\hat{}t\, Hx + x\hat{}t\, H\hat{}t\, Hx$$

The update variable obtained using Major Maximization will be;

$$(x_{k+1}) = x_k + \frac{1}{\alpha} H^t(y - Hx_k)$$

Pseudo Code for Reconstruction
For max number of iteration
Return x for minimal OBJ

$$\Rightarrow \boldsymbol{OBJ(x) = argmin_x\, ||y - Hx||^2{}_2 + \lambda||x||_1}$$

$$\boldsymbol{x_{k+1} = soft(x_k + \tfrac{1}{\alpha}H^T(y - Hx_k), \tfrac{\lambda}{2\alpha})}$$

End

Soft Thresholding

It is a function which is defined as;

$$soft(x, T) = \begin{cases} x + T & x \le T \\ 0 & |x| \le T \\ x - T & x \ge T \end{cases}$$

And it is used as x = soft (y,λ/2).

C. Denoising

The reconstruction model above does not consider the presence of various noises and distortion in the signals. The main objective of this module is to focus on the denoising of these EEG signals using Lasso Algorithm. Here the signal that is to be denoised, can be either a reconstructed signal or the directly measured original EEG signal.

Lasso is also an algorithm which is Used for solving minimization problems and can be applied on the above objective function. We identified two major disturbances that might be present. Additive Gaussian Noise which is used to simulate the thermal noise introduced by equipment and components. The second noise is Additive Laplacian Noise. In Communication systems it is used to model the interference, here the same noise model can be used to simulate the interference in the weak EEG signals.

$$minimize\ ||y - Hx||_2{}^2 + \lambda||x||_1$$

Here the update x can be computed as;

$$x_{k+1} = (H^T H^* + \rho I)^{-1}(H^T y + \rho(z^k - u^k))$$

$$z_{k+1} = soft(x_{k+1} + u_k, \frac{\lambda}{\rho})$$

$$u_{k+1} = u_k + x_{k+1} + z_{k+1}$$

Pseudo Code for Denoising
For max number of iterations
Update following variable (x,z,u)

$$\boldsymbol{x_{k+1} = (H^T H^* + \rho I)^{-1}(H^T y + \rho(z^k - u^k))}$$

$$\boldsymbol{z_{k+1} = soft(x_{k+1} + u_k, \frac{\lambda}{\rho})}$$

$$\boldsymbol{u_{k+1} = u_k + x_{k+1} + z_{k+1}}$$

Compute objective function and minimize it

$$\boldsymbol{minimize\ ||y - Hx||_2{}^2 + \lambda||x||_1}$$

End

IV. Results

Fig. 2 Shows the Frequency response of the original EEG Signal which we used to compare it with reconstructed signal further in the section.

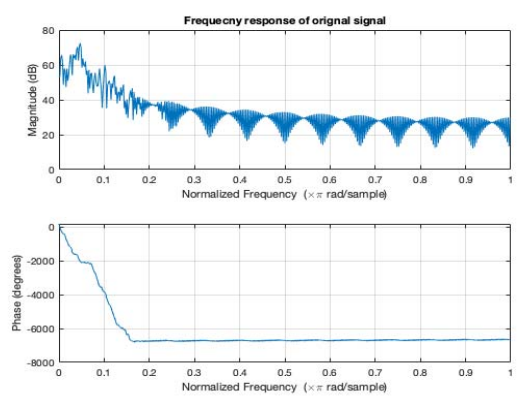

Fig. 2. Frequency response of original signal

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Below Fig. 3, shows the reconstructed signal using 25% of the original samples. We can interpret that it is similar to the original signal.

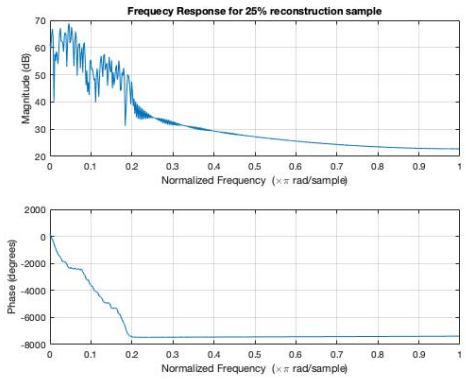

Fig. 3. Frequency response for 25% reconstruction samples

Fig. 4, shows the reconstructed signal using 25% samples with standardization for a better output , however we can observe delay, yet the tracing is alike the original signal.

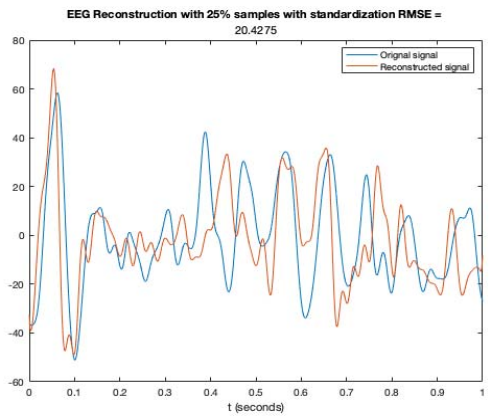

Fig. 4. EEG Reconstruction with 25% samples with standardization

Comparing Fig. 4 and Fig. 5, we can see that standardization provides a better output , compared to non-standardized data.

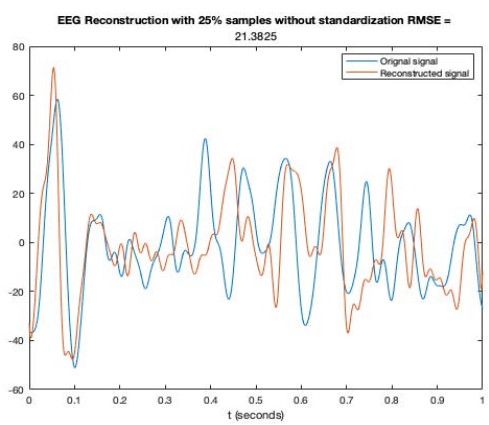

Fig. 5. EEG reconstruction with 25% samples without standardization

Fig. 6, shows the original reconstructed signal before interpolation. We can conclude that our algorithm, reconstructs on point results.

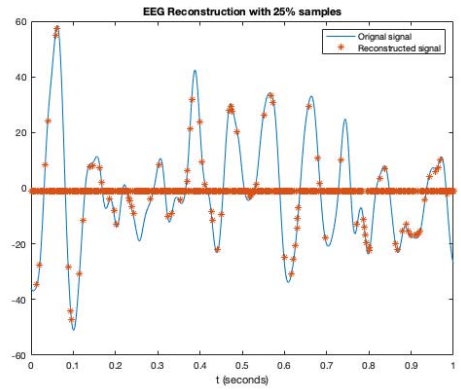

Fig. 6. EEG Reconstruction with 25% samples – sparse signal

Below Fig. 7, shows the reconstructed signal using 10% of the original samples. We can interpret that it is alike to the original signal.

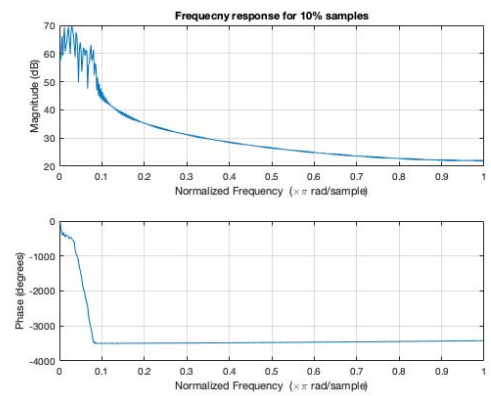

Fig. 7. Frequency response for 10% reconstruction samples

Fig. 8, shows successful reconstructed signal using 10% samples, yet we still observe delay, yet the tracing is alike the original signal.

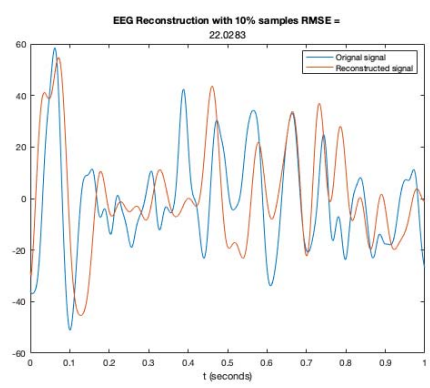

Fig. 8. EEG reconstruction with 10% samples

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Fig. 9 and Fig. 10 shows perfect denoising of the noisy signal , which is indistinguishable from the original clean signal.

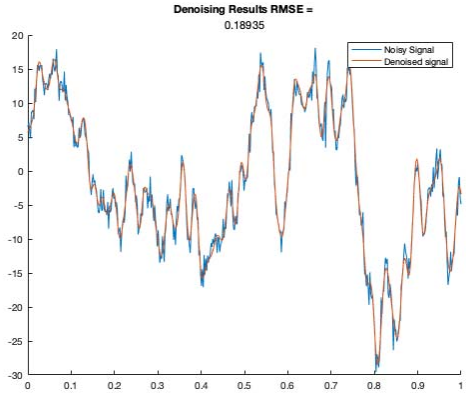

Fig. 9. Denoising of EEG signal

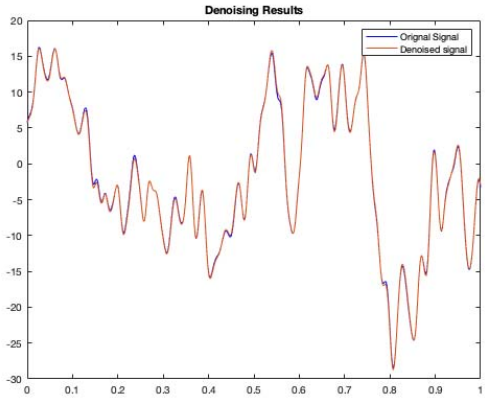

Fig. 10. Original signal and denoised signal

Comparing Fig. 13 and Fig. 2, we can see that our algorithm has achieved almost perfect result for denoising. Fig. 12 shows the frequency response of a noisy signal we generated, which was cleaned by our algorithm

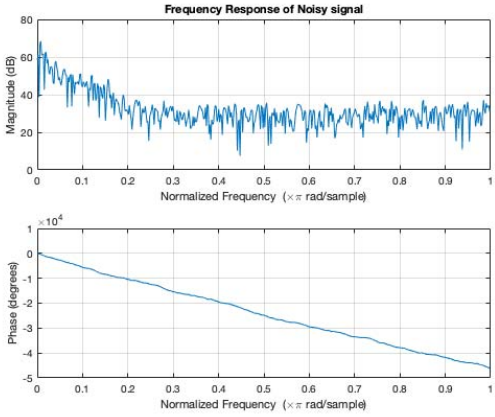

Fig. 12. Frequency response of noisy signal

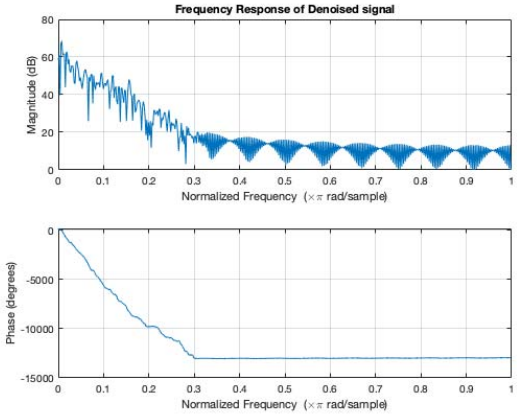

Fig. 13. Frequency response of denoised signal

V. Conclusion

This project performs exhaustive research and work on the new trend in signal processing. This project develops and performs ADMM based signal restoration for biomedical signal processing. Compressed sensing is surely an innovative technique for acquiring and restoring signals and when implemented using ADMM it delivers promising output which is both computationally efficient and accurate. The Denoising also takes assistance from Lasso based on ADMM. The model shows robustness to all the noise that were employed in the simulation environment for developing practical scenario.

The proposed model didn't deliver ideal RMSE value however, the RMSE is not a correct evaluation measure for the simulation, it's close to. The ideal sparse reconstruction was obtained, however an interpolation function distorts the resulted output. To avoid the use of interpolators, Discrete Cosine Transforms were used but that resulted in similar RMSE and reconstruction outcomes. The ADMM solver in Lasso is able to perform perfect denoising of the EEG signal. The RMSE values are in the desired range of 0.1 to 0.2. The Model comprises thermal noise and interference. Even in denoising model standardization of signal helped in achieving desired RMSE and qualitative output.

The ADMM offers a very powerful approach to the various optimization, reconstruction, compression and various application essential to the real world operations.

VI. Future Scope

We further plan to work on creating a better reconstruction algorithm and also work to design an interpolator which won't introduce nonlinear phase shift. Furthermore, the concepts of deep learning and convex optimization can be explored and introduced into the system model. These algorithms in proposed projects aren't limited to biomedical application. We further plan to apply these algorithms for financial engineering and for political theories for diverse practical application. The EEG is one of many biomedical signals. Hence this project can be further employed on various biomedical and general signals.

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